

Description

A woman arrives to the NIST Hospital's Emergency Department in active labor, and delivers in the emergency department at 7:30 AM on February 1, 2025. The newborn has a Congenital diaphragmatic hernia (CDH), which is entered as a working diagnosis, and has suffered Meconium aspiration syndrome (MAS), which is entered as final diagnoses as they were identified upon delivery. At 7:35, a clerical assistant registers the newborn as a patient. She records the patient's date of birth, the height and weight taken at birth, race, ethnicity, residence, insurance information (PlanID123 from MyInsuranceCompany). There is no patient name for the newborn at the time of birth.

Southern Midwest Medical Center reports syndromic surveillance data to the City Health Department (NIST PH SS FAC 1). At 7:45 AM on February 1, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits a Registration message about this encounter. The placeholder patient name and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Comments

Dates and times are provided in this test case to illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the Test Case with an HIT Module, only date and time format will be validated within tester submitted test data.

Pre-condition

No Pre-Condition

Post-Condition

No Post-Condition

Test Objectives

This test step examines an HIT Module's ability to create an ADT^A04 Registration message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019, conformance requirements.

The following test objectives and variants are supported by this test case:

- Setting is Emergency Department with Inpatient Admission.
- Chief complaint is captured as free-text.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identified manner is required for this Test Step.
 - Multiple Races.
 - Pediatric case where age is measured in months.
 - Inclusion of Admitting Diagnoses.
 - Insurance is unknown

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
P2401	Meconium aspiration with respiratory symptoms	I10C	P2401^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
206292002	Meconium aspiration syndrome (disorder)	SCT	206292002^Meconium aspiration syndrome (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
P24.01	Type III occipital condyle fracture, initial encounter for closed fracture	I10C	P24.01^Type III occipital condyle fracture, initial encounter for closed fracture^I10C	Yes
P2401	Type III occipital condyle fracture, initial encounter for closed fracture	I10	P2401^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
P24.01	Type III occipital condyle fracture, initial encounter for closed fracture	I10	P24.01^Type III occipital condyle fracture, initial encounter for closed fracture^I10	No
276533002	Neonatal aspiration syndrome (disorder)	SCT	276533002^Neonatal aspiration syndrome (disorder)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
Q790	Congenital diaphragmatic hernia	I10C	Q790^Congenital diaphragmatic hernia^I10C	Yes
17190001	Congenital diaphragmatic hernia (disorder)	SCT	17190001^Congenital diaphragmatic hernia (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
Q79.0	Congenital diaphragmatic hernia	I10C	Q79.0^Congenital diaphragmatic hernia^I10C	Yes
Q790	Congenital diaphragmatic hernia	I10	Q790^Congenital diaphragmatic hernia^I10	No
Q79.0	Congenital diaphragmatic hernia	I10	Q79.0^Congenital diaphragmatic hernia^I10	No
88386004	Congenital anomaly of diaphragm (disorder)	SCT	88386004^Congenital anomaly of diaphragm (disorder)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues an error.

Notes

Messages in this test case will be used in a Manual Test Step under functional requirement testing.